

Zakhar Shumaylov

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LinkedIn: linkedin.com/in/zakshumaylov

Scholar: scholar.google.com/citations?user=EKV1lrAAAAAJ

Education	University of Cambridge PhD in Mathematics of Information Thesis: “On Strengths and Limitations of Data-Driven Approaches in Inverse Problems.” Supervised by: Prof Carola-Bibiane Schönlieb. Awarded the <i>Trinity Henry Barlow Scholarship</i> (£81,000) at Christs College. Funded by Christs College Warwick Postgraduate Studentship (£15,000) and CCIMI (£50,000).	Cambridge, UK 2022 – 2026
	University of Cambridge Mathematics BA/MMath (1st Class/Distinction) Awarded the <i>Cambridge Trust Scholarship</i> (£40,000) to read Mathematics at Churchill College. Courses included: Quantum Field Theory, General Relativity, Statistical Field Theory, Black Holes, Cosmology.	Cambridge, UK 2018 – 2022
	Brighton College A-Level (5A*), STEP 2,3 (S,S)	Brighton, UK 2016 – 2018
	Governor’s Physics and Mathematics Lyceum 30 Year 9 - Year 11 (4.53/5)	St-Petersburg, Russia 2013 – 2016

Publications	 [17] AI models collapse when trained on recursively generated data Z. Shumaylov* , I. Shumailov*, Y. Zhao, Y. Gal, N. Papernot, R. Anderson NATURE, 2024 Selected as cover and ranked 18th / 300k of articles . Covered on the front page of New York Times . One of the most influential articles of the year per State of AI report . Publicity: New Scientist ; Independent ; The Atlantic ; MIT tech ; Financial Times ; New York Times ; Wall Street Journal ; Bloomberg ; The Register ; AI Magazine ; Cosmos	
	[16] When is a System Discoverable from Data? Discovery Requires Chaos Z. Shumaylov* , P. Zaika*, P. Scholl, G. Kutyniok, L. Horesh, C. Schönlieb UNDER REVIEW, 2025	
	[15] Generalized Lie Symmetries in Physics-Informed Neural Operators Z. Shumaylov* , A. X. Wang*, P. Zaika, F. Sherry, C. Schönlieb ORAL AT SCML 2025 ; — ORAL & BEST PAPER RUNNER-UP AT COLT TASC 2025	
	[14] Learning Regularization Functionals for Inverse Problems: A Comparative Study J. Hertrich, H. S. Wong, A. Denker, S. Ducotterd, Z. Fang, M. Haltmeier, Z. Kereta, E. Kobler, O. Leong, M.S. Salehi, C. B. Schönlieb, J. Schwab, Z. Shumaylov , J. Sulam, G. S. Wache, M. Zach, Y. Zhang, M. J. Ehrhardt, S. Neumayer UNDER REVIEW, 2025	

- [13] **Score-based pullback Riemannian geometry** 
W. Diepeveen, G. Batzolis, **Z. Shumaylov**, C. Schönlieb
ICML, 2025
- [12] **Lie Algebra Canonicalization: Equivariant Neural Operators under arbitrary Lie Groups** 
Z. Shumaylov^{*}, P. Zaika^{*}, J. Rowbottom, F. Sherry, M. Weber, C. Schönlieb
ICLR, 2025
- [11] **On Information Geometry and Iterative Optimization in Model Compression: Operator Factorization** 
Z. Shumaylov, V. Tsiaras, Y. Stylianou
UNDER REVIEW, 2025
- [10] **Symplectic Neural Flows for Modeling and Discovery** 
P. Canizares, D. Murari, C. Schönlieb, F. Sherry, **Z. Shumaylov**
UNDER REVIEW, 2024
- [9] **Hamiltonian Matching for Symplectic Neural Integrators** 
P. Canizares, D. Murari, C. Schönlieb, F. Sherry, **Z. Shumaylov**
ORAL AT **NEURIPS**, 2024
- [8] **Benchmarking Learned Algorithms for Computed Tomography Image Reconstruction Tasks** 
M. Kiss, A. Biguri, **Z. Shumaylov**, F. Sherry, J. Batenburg, C. Schönlieb, F. Lucka
APPLIED MATHEMATICS FOR MODERN CHALLENGES, 2025
- [7] **Data-driven approaches to inverse problems** 
C. Schönlieb, **Z. Shumaylov**
CIME 2023 NOTES, 2025
- [6] **Weakly Convex Regularisers for Inverse Problems: Convergence of Critical Points & Primal-Dual Optimisation** 
Z. Shumaylov, J. Budd, S. Mukherjee, C. Schönlieb
ICML, 2024
- [5] **Data-Driven Convex Regularizers for Inverse Problems** 
S. Mukherjee, S. Dittmer, **Z. Shumaylov**, S. Lunz, O. Öktem, C. Schönlieb
ORAL AT IEEE ICASSP, 2024
- [4] **Provably Convergent Data-Driven Convex-Nonconvex Regularization** 
Z. Shumaylov, J. Budd, S. Mukherjee, C. Schönlieb
ORAL AT **NEURIPS WORKSHOP ON DEEP LEARNING AND INVERSE PROBLEMS**, 2023
- [3] **Quantum Initial Conditions for Curved Inflating Universes** 
Z. Shumaylov^{*}, M. Letey^{*}, F. Agocs, W. Handley, M. Hobson, A. Lasenby
PHYSICAL REVIEW D, 2024

[2] **Primordial power spectra from k-inflation with curvature**

Z. Shumaylov, W. Handley

PHYSICAL REVIEW D, 2022



[1] **Manipulating SGD with data ordering attacks**

I. Shumailov, **Z. Shumaylov**, D. Kazhdan, Y. Zhao, N. Papernot, M. Erdogdu, R. Anderson

NEURIPS, 2021



Experience	Google DeepMind	Mountain View, USA
	<i>Student Researcher</i>	08/25 – 01/26
	Research on deep learning for inverse optical lithography.	
	Apple	Cambridge, UK
	<i>ML Research Intern</i>	12/24 – 08/25
	ML research on model compression.	
	Apple	Cambridge, UK
	<i>ML Research Intern</i>	06/24 – 09/24
	ML research on model compression using tensor networks.	
	University of Cambridge, DAMTP	Cambridge, UK
	<i>Supervisor for Undergraduates and Postgraduates</i>	10/22 – Present
	Supervising undergraduate students in a variety of courses.	
	(2022/2023): Part IA Vectors and Matrices: 18 students (48h);	
	(2023/2024): Summer Project Supervision: 2 students;	
	(2024/2025): MPhil and Summer Project Supervision: 2 + 2 students.	
	GSK	Cambridge, UK
	<i>Project Collaboration</i>	06/22 – 09/22
	Project collaboration on “Self-discovery of mechanistic equations for a data-driven smart simulator” as part of CMI programme with Dr Matthieu Duvinage.	
	Ryff AI	Cambridge, UK
	<i>Summer Research Intern</i>	07/22 – 09/22
	Work under supervision of Dr Mike Roberts. During the internship I worked on the problem of unsupervised video motion segmentation. During the project, I used variational and learned methods from the optical flow literature for foreground-background separation using motion signals.	
	University of Cambridge, Institute of Astronomy	Cambridge, UK
	<i>Summer Internship Programme</i>	08/21 – 09/21
	Work under supervision of Dr Amy Bonsor (IoA): “Gas disk imaging around white dwarves”. During the internship I investigated gas disk light curve imaging around white dwarves, by modelling gas geometry. Funded by the Institute of Astronomy.	
	University of Cambridge, KICC	Cambridge, UK
	<i>Summer Research Intern</i>	06/21 – 08/21
	Work under supervision of Dr Will Handley (KICC): “Primordial power spectra from k-inflation with curvature”. During the internship I investigated the problem of interplay between inflationary sound speed and primordial curvature using analytical approximations. Funded by the CMP .	

University of Cambridge, DAMTP	Cambridge, UK
<i>Summer Research Assistant</i>	06/20 – 09/20
Work under supervision of Prof Carola Schonlieb (DAMTP), Prof Ozan Oktem (KTH) and Prof Par Kurlberg (KTH): “3DEM: Representation of atomic models”. During the internship I investigated the problem of protein fitting inside of atomic volumes acquired via cryo electron microscopy. During the project I used learned techniques and variational methods to obtain protein reconstructions. Funded by the CSRIM .	
University of Cambridge, DAMTP	Cambridge, UK
<i>Summer Research Assistant</i>	06/19 – 09/19
Work under supervision of Prof Carola Schonlieb (DAMTP). During the internship I worked primarily in the field of inverse problems. In particular, I researched how Deep Learning can be used to help solve physics-based inverse imaging problems. This led to a joint work “Learned convex regularizers for inverse problems”. Funded by the CSRIM and the Tizard Fund .	
Cambridge Coding Academy	Cambridge, UK
<i>Teaching Assistant</i>	07/18
Supporting and leading coding sessions of the ‘Coding++’ course, covering the basics of AI using python and the pygame library.	
Brighton College	Brighton, UK
<i>After-school Teaching Assistant</i>	09/17 – 06/18
Tutoring Year 9 - Year 13 students during after-school Mathematics classes.	
University of Sussex	Brighton, UK
<i>Research Assistant to Professor Madzvamuse</i>	07/17 – 08/17
I reviewed and extended the one-dimensional cell model of Shenoy(2016) by modelling cell contractility and strain with partial differential equations in Matlab.	

Subject Olympiads	British Physics Olympiad Round 2 (Gold Award, Top 15)	2018
	Invited to University of Oxford Training Camp to compete for UK IPhO team.	
	British Astronomy and Astrophysics Olympiad (Gold Award)	2018
	British Physics Olympiad Round I (Gold Award, Top 50)	2017
	British Mathematics Olympiad Round I (Certificate of Distinction)	2017
	British Physics Olympiad Round I & AS Physics Challenge (Gold Awards)	2016
	Senior Mathematics Challenge (Gold Award, 100%)	2016
	School Mathematics Olympiad (Winner of inter-school team challenge)	2016
	Russian Computer Science & Physics Olympiads (Winner of district challenges)	2015
Awards & Fellowships	Russian Computer Science Olympiad (Winner of district challenge)	2014
	Trinity Henry Barlow Scholarship (£81,000)	2022
	Awarded to pursue PhD in Mathematics of Information at University of Cambridge.	
	Cambridge Christs Bursary (£15,000)	2022
	Awarded to pursue PhD in Mathematics of Information at University of Cambridge.	

	CCIMI (£50,000)	2022
	Awarded to pursue PhD in Mathematics of Information at University of Cambridge.	
	C.I.M.E. full grant (€ 1,000)	2023
	Awarded to attend CIME school on 'Machine Learning: From Data to Mathematical Understanding'.	
	Churchill College Prize Scholarship (£120)	2021
	Awarded in recognition of excellent academic performance.	
	Churchill College Honorary Scholarship (£100)	2020
	Awarded in recognition of excellent academic performance.	
	Churchill College Prize Scholarship (£120)	2019
	Awarded in recognition of excellent academic performance.	
	Cambridge Trust Scholarship (£40,000)	2018
	Awarded to read Mathematics at University of Cambridge.	
	Brighton College Governors Award for Independent Study (£500)	2018
	For a piece of work outside of the A-Level curriculum.	
	Brighton College Physics Prize: Bayliss-Smith prize	2018
	Prize to recognise sustained excellence and scientific endeavor.	
	Brighton College Science Essay Competition	2018
	Winning essay: "The Tale of Cell Modelling".	
	Brighton College Science Prize: Newton's Cup	2017
	Prize to recognise sustained excellence and scientific endeavor.	
	Brighton College Science Essay Competition	2017
	Winning essay: "Brief History of Exoplanets".	
Invited Talks	NeurIPS @ Cambridge (Cambridge, UK):	
	Invited to present on "When is a System Discoverable from Data?"	
	Machine Learning Journal Club (MLJC) at Gatsby UCL (London, UK):	
	Invited to present on "The Future of Synthetic Data: Model Collapse and Equivariant Neural Operators."	
	Workshop on Lie Groups and Symmetry at The Alan Turing Institute (London, UK):	
	Invited to present on "Symmetries in Neural O/PDE Solvers."	
	Maths4DL Conference on Inverse Problems and Deep Learning (Bath, UK):	
	Invited to present on "Symmetries in Neural PDE Solvers."	
	BAMC 2025 (Exeter, UK):	
	Invited to present on "Convergent Data-Driven Regularisation in Inverse Problems."	
	NUS (Singapore):	
	Invited to present on "Symmetries in Neural O/PDE Solvers."	
	Harvard University (Boston, USA):	
	Invited to present on "Symmetries in Neural O/PDE Solvers."	

Christs College (Cambridge, UK):

Invited to present on “AI Models collapse when trained on recursively generated data” as part of ERSS series.

TU Berlin (Berlin, Germany):

Invited to present on “AI Models collapse when trained on recursively generated data.”

Tübingen AI Center (Tübingen, Germany):

Invited to present on “The Future of Synthetic Data: Model Collapse and Equivariant Neural Operators.”

Oberwolfach workshop on “Deep Learning for PDE-based Inverse Problems” (Oberwolfach, Germany):

Invited to present on “Lie Algebra Canonicalization: Equivariant Neural Operators under arbitrary Lie Groups.”

European Congress of Mathematics 2024 (Seville, Spain):

Invited to present on “Weakly convex regularizers in inverse problems.”

KTH SciML workshop (Stockholm, Sweden):

Invited to present on “Weakly convex regularizers in inverse problems.”

AI Precision Health Institute (Hawaii, USA):

Invited to present on “What happens if we use synthetic data without any curation.”

SIAM Imaging 2024 (Atlanta, USA):

Invited to present on “Weakly convex regularizers in inverse problems.”

IEEE ICASSP 2024 (Seoul, South Korea):

Invited to present on “Data-Driven Convex Regularizers for Inverse Problems.”

NeurIPS @ Cambridge (Cambridge, UK):

Presented on “The Curse Of Recursion: Generated Data Makes Models Forget.”

Workshop: Integrating acquisition and AI in tomography (Leiden, Netherlands):

Presented on “Learned reconstruction methods in inverse problems.”

ICIAM 2023 (Tokyo, Japan):

Invited to present on “Learned weakly convex regularizers in inverse problems.”

C.I.M.E. School on ‘Machine Learning: From Data to Mathematical Understanding’ (Cetraro, Italy):

Received full grant and prepared lecture notes to be published in the C.I.M.E. Springer series.

Service

Reviewing for Conferences: [ICML](#), [ICLR](#), [NeurIPS](#), [IEEE ICASSP](#), [AAAI](#)

Reviewing for Workshops: [SLLM](#), [NeurReps](#), [GRaM](#)

Reviewing for Journals: [IMA Journal of Numerical Analysis](#), [Philosophical Transactions of the Royal Society A](#), [IEEE Transactions on Computational Imaging](#), [Foundations of Data Science](#), [ACM Transactions on Intelligent Systems and Technology](#), [Journal of Computational Physics](#), [Numerical Functional Analysis and Optimization](#)

Organization: [NeurIPS @ Cambridge meetups](#) (2023, 2024).

Leadership Positions: Treasurer and Membership officer at Cambridge University Astronomical Society; Deputy Head of School House at Brighton College; Founder and President of Brighton College STEM Society; Leader of the House Chess Team at Brighton College.

Technical Skills

Programming Languages: Python, C

Software Packages: pyTorch, jax, odl, Matlab

Other Tools: Maple, Mathematica, LaTeX

OS & Computing: Linux, MacOS, unix, bash, slurm, HPC, vim